Turn-Level Transformation: Methodology

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Purpose and scope

The transformation splits each article in the turn set into turns: one row per stretch of speech or narration by one speaker. A member's own words can then be pulled out and put in date order as an oral history of each act, as Harrison, Askin and Hagtvedt (2023) did. The rules run in nine stages of Python, with no language model.

Input. The 9,316 records of the turn set: the cleaned article records with high date confidence and no exclusion.

Output. 132,835 turns: 86,870 of speech, 44,065 of narration and 1,900 second-hand. Each turn records its speaker, its confidence and the rule that named the speaker, with the attribution or label that triggered it where there is one.

Voices.

1. **Band voice:** the act's members at the time of the nomination speaking in their own words, singly or as a group. In prose it is a direct quotation; in a Q&A, the member's answer; in a first-person account, the account itself.
2. **Public narrative:** everything else, that is, the journalist's narration, the interviewer's questions and quotations from other people, such as producers, managers and relatives.
3. **Second-hand:** the journalist reporting a member's words without quotation marks, as in the invented sentence "Adam says they have been practising for 10 hours every day". The turn is the journalist's, with the member recorded as the reported member, and it is never band voice.
4. **Other-era member:** a member who joined after the nomination or had left before it. Their words are named as theirs but kept apart from band voice.
5. **Unknown:** a quotation whose speaker cannot be named on evidence. Unknown is never band voice.

Choices the method rests on.

1. Harrison et al. ordered each act's interviews by date, so reliable dates are part of the method: only records with high date confidence go forward.
2. Coding happens at the level of the turn, not the record.
3. Paraphrase and indirect speech are not band voice.

4. Turns are built by rules, not by a language model, so that every attribution follows a stated rule.
5. Public narrative is kept as its own layer, beside band voice.

The stages and their scripts.

1. Stages 1 and 2, normalise and route: `turns_v7.py`, version 2.
2. Stage 3, speakers in Q&As and first-person accounts: `turns_v7_stage3.py`, version 3.
3. Stages 4 to 9, quotations and their speakers in prose, then turns, interview dates and confidence for every record: `turns_v7_stage4to9.py`, version 1.3.

Each script reads the files of the one before, checks that they are the expected versions, never changes them, and writes new ones.

Principles

Seven principles hold at every stage.

1. **The raw text is never changed.** Stage 1 swaps characters one for one, so every position in the normalised text matches the raw text, and any turn can be traced to the exact place it came from.
2. **Every character belongs to exactly one span.** Joining a record's spans in order rebuilds its text exactly. This round trip is checked on every run, so no text is lost or counted twice.
3. **Quotation marks are not automatically speech.** Song titles, album names and scare quotes sit in them too.
4. **Unknown is better than wrong.** A wrong attribution corrupts the data without anyone noticing, while an unknown only reduces coverage. So a speaker is named only on evidence, and unknown is never band voice.
5. **Attribution is judged across the whole article.** Who is speaking depends on what came before, not on the quotation alone.
6. **Every attribution records its evidence:** the rule that named the speaker and, where there is one, the attribution or label that triggered it, so that each rule can be measured on its own.
7. **Rules are measured, not trusted.** The confidence levels were checked against blind coding, by AI agents during development and by a person on held-out records.

Spans and turns

A span is one piece of an article's text, cut at a start and an end, in which everything is the same kind of text: someone's spoken words, the journalist's own words, a clause such as "she says", or a speaker label such as "Q:". A turn is one or more spans by the same

speaker, read as one contribution.

The invented passage *"We were exhausted," she says. "We had been touring for months."* has three spans but one turn:

Span	Text	Type
1	"We were exhausted,"	Speech
2	she says.	Attribution
3	"We had been touring for months."	Speech

Spans 1 and 3 have the same speaker, so they form one turn. Span 2 is kept on that turn as the evidence for who spoke.

Name lists

The stages that name speakers rely on eight lists. They were first built from a scan of all 52 acts, built again after the line-ups were checked, and every entry the script was unsure of was decided by hand. The lists are written into the stage 3 and stage 4 to 9 scripts, and the full lists are in `name_lists_review.xlsx` in the repository.

1. **Members:** 247 people across the 52 acts, each with their era, the pronoun used for them and, where known, the years they joined and left. The 149 nomination-era members (the line-up at the time of the nomination, or the solo artist) are the band, and only their words are band voice. The other 98 joined after the nomination or had left before it. They are named as speakers but kept out of band voice, as other-era members.
2. **Aliases:** 154 other forms of 95 members' names in 41 acts: short and full forms, stage names and nicknames, birth or real names and spelling variants. "Phil Oakey" is Philip Oakey and "Mags" is Magne Furuholmen.
3. **Duplicate entries:** three people listed under two names each resolve to one speaker: Sheila E. and Sheila Escovedo, Tone Loc and Anthony Terrell Smith, and Terence Trent D'Arby and Sananda Maitreya.
4. **Non-members who share a name with a member:** 88 relatives and namesakes across 35 acts, such as John Lennon in Julian Lennon's records and Pete Escovedo in Sheila E.'s. A list like this is never complete, so the scripts find relatives and namesakes in each record as they read it (stages 3 and 5). In stages 4 to 9 the list also stops an alias from making a non-member's name: "John Charles Julian Lennon" would otherwise make "John Lennon" read as Julian Lennon.

5. **Stoplist:** 52 capitalised words, such as Although, Captain and Young, that can stand before a surname but are not first names, so that "Although Vandross" never becomes a name.
6. **Names shared within an act:** 22 first names, surnames or initials shared by two or more people in 14 acts, such as Knopfler in Dire Straits and JC (Joanne Catherall and Jo Callis) in The Human League. The scripts settle them with the general rules for shared names (stages 3 and 5). One surname is never used alone: in Big Country's records, "the Stones" usually means the Rolling Stones.
7. **Q&A labels:** 49 labels checked in 23 acts, such as LAUPER for Cyndi Lauper and PO for Philip Oakey. One label can name different people in different acts, so a label is matched only against the act's own people.
8. **Labels that are not speakers:** a fixed list of 392 labels, written into the stage 2 and 3 scripts, that never count as speakers, except that stage 3 reads initials on it, such as PR, as a speaker's label when someone named in the record has those initials. The frequent ones are photo credits, release details, review furniture, links and buttons, and page or event details.

The line-up check. Each of the 30 line-ups of groups and duos was checked against Wikipedia's member list and at least one other source, such as AllMusic, album credits, band or fan sites, or the corpus itself. 22 were right as they stood, 5 needed changes and 3 had members whose timing was unclear. The 23 member entries at issue, 14 changes and 9 uncertain members, were decided by hand, and the 10 borderline calls are flagged in the review workbook.

Two rules from the check of the lists.

1. A speaker label counts as a member only where the record also names that member in full, with two narrow exceptions (stage 3); where two people could own a label, the one named in the title or introduction wins. In one radio interview, "Harris:" was the host, not the band's Peter Harris.
2. When two people who share a name belong to different eras, a bare shared name counts only if an earlier full name settles it. Otherwise it is ambiguous, at low confidence, and kept out of band voice.

Stage 1: normalise

Stage 1 swaps characters one for one, so the text keeps its length and every position still points into the raw text, which is never overwritten. It makes seven kinds of swap, in this order:

1. Curly and other double quotation marks become straight double quotes (").

2. Curly and other single quotation marks, primes, accents used as apostrophes and single backticks become straight single quotes (').
3. Non-breaking and other unusual spaces become ordinary spaces.
4. Unicode line and paragraph separators become line breaks.
5. Guillemets (« ») become double quotes, but only where they pair within one paragraph.
6. A double quote straight after 7, 10 or 12, as in 12" single, becomes an inch sign (″), since it is a measurement, not a quotation mark. This does not apply when the number follows a digit, a full stop, a slash or a comma. 12'' becomes ″ and a space.
7. Two-character quotation marks (``x'' and ''x'') that pair within a paragraph each become a double quote and a space. A leftover `` becomes an opening quote, and an unpaired '' is changed only in records with at least two pairs.

Nothing else changes: there is no Unicode normalisation, and dashes and spacing are left alone. A list of allowed swaps and a length check guard every record, the run stops before writing if any record fails, and the files are read back after writing.

The swaps are needed because a Q&A label indented with a non-breaking space is invisible to a pattern that expects an ordinary space, and an inch mark would unbalance its paragraph's quotation marks.

Stage 2: route

Stage 2 routes each record by its format, because in Q&As and first-person accounts the format, not quotation marks, shows who is speaking. Read as prose, a member's answers would become the journalist's narration. The tests run in a fixed order, each only while the record is still prose, and the first that passes decides the route.

Route	Records
Prose	7,814
Unlabelled Q&A	652
Labelled Q&A	631
First-person account	170
Forum thread (not processed)	49

1. **Forum threads.** A record whose page type is a forum thread becomes one segment and goes no further. One record of machine-generated text is also left out from stage 3 on, so 9,266 records are processed.

2. **Labelled Q&As.** Labels at the start of a line are collected in the forms in use: NPR's "NAME, ROLE:", "Name, role:", a colon, a dash, a bracketed name, initials, and bare names that recur. Labels over 40 characters, 392 known metadata labels, clauses such as "He added", pronouns and cues for sung parts are dropped, and one person's labels are merged. The record is a labelled Q&A when these lines pass one of eight tests. The main test needs six or more label lines from at least two labels, at least two changes of speaker, the two commonest labels on at least 60% of the lines, a median line of at least 40 characters for one label, and enough of the lines reading as speech. The other tests cover radio transcripts, panels of many speakers, short interviews, interviews in which only the questions or only the answers are labelled, and a single labelled speaker. A pass is cancelled when narration fills more than half of the labelled stretch. Four rarer layouts are caught after these: labels inside paragraphs, 'Name: "quotation"' lines, lecture transcripts and caption transcripts.
3. **Unlabelled Q&As.** A question is a paragraph of 8 to 600 characters that ends with a question mark, does not open with a quotation mark, is less than 70% capitals and is not a list item. An answer has at least 15 characters and no attribution in the third person. The record passes on one of five tests. The main test needs four or more questions with their answers, covering at least a quarter of the paragraphs, with at least half of the answers in the first person. The others weigh how often the questions say "you", a Q&A title, the number of distinct questions and the length of the questions.
4. **First-person accounts.** First-person words are counted per 1,000 words outside quotation marks, and attributions and quotation marks per 1,000 words of the whole text, in the record's own quotation convention. A record is a first-person account when it fits one of six patterns of at least 150 words (100 for interview highlights): headings that name two or more speakers; three or more topic headings under a title that announces them; "Interview highlights"; one speaker throughout, with at most 1.5 attributions and 8 quotation marks and at least 35 first-person words (45 counting "we") per 1,000 words, and not lyrics; an "X on Y" title; or a title that says so, in which case quoted words count too.
5. **Prose:** everything else.

The exact thresholds of every test are in `turns_v7.py`. A check by hand of the routes and block starts of 54 records shaped these rules (see Validation).

Blocks

Each record's segments cover its text exactly once: a prose introduction, the interview or account as one or more blocks, prose gaps between blocks, and a prose tail. Every prose segment goes on to stages 4 to 9, so a quotation in an introduction can still become speech.

1. **Labelled Q&As.** The block starts at the first label that is used twice, is in NPR style or looks like a person's name, leaving out the headline. It moves back to an earlier label line when at most two unlabelled paragraphs lie between them, and takes in a question just before it. It ends at the last label plus up to eight paragraphs, stopping earlier at page furniture, a sign-off or narration. A paragraph inside the block that quotes a labelled speaker in the third person becomes prose.
2. **Unlabelled Q&As.** The block is built from the questions and answers, with neighbouring ones merged. It can start in the middle of a paragraph, at a question that follows 250 or more characters of narration.
3. **First-person accounts.** The block starts at the first first-person paragraph or heading, with the heading just above it, and ends at page furniture or the writer's biography.
4. **In every block,** page furniture is left out and NPR's copyright line is cut. Trailing clutter is trimmed from the last block, and a lone closing question is dropped when there are several blocks.

Flags

Some records are flagged without being excluded: not in English (28), a page whose interview is audio or video (212), lost line breaks (54) and under 300 characters (10). Each record also stores its route, the pattern that decided it, its segments and the evidence for the route, so accuracy can be measured route by route.

Stage 3: speakers in Q&As and first-person accounts

In these records the format shows where one speaker stops and the next begins, so the work is deciding whose each label, answer or account is. Quotation marks inside an answer or an account are the speaker reporting someone else's words: they stay inside the turn, which counts its embedded quotes.

Matching a name

1. Names are matched regardless of case and accents, with a space or a hyphen between words. Honorifics (Mr, Ms, Mrs, Dr, Miss, Sir) are stripped from labels, so NPR's "Ms. CHAPMAN" is Tracy Chapman.
2. A full name or an alias counts on its own. A first name, a surname or initials (two to four capitals) count only for a member named in full in the title or introduction, or before the label is first used. For a first name or surname that one member alone has and no non-member shares, two exceptions give medium confidence: the other half of the name is in the title or introduction, or the label has four or more letters and the title or introduction names the act.

3. Where members share the name, the one named in full gets the label (high). If two or more are named in full, the one in the title or introduction wins (medium); otherwise the speaker is "band member, unspecified" (medium) when all of them are nomination-era members, "other-era member, unspecified" (medium) when all are from other eras, and ambiguous (low) when the eras are mixed. If none of them is named in full, the label is another person's (medium).
4. Relatives and namesakes are found in each record, not taken from the list: any non-member named in full. If one of them fits a short-form label and is in the title or introduction while the member is not, the label is theirs (other person, medium). If neither is there, the label is ambiguous (low). A relative who shares the member's surname lowers the confidence by one level.
5. A stoplist word never starts a name; as a label, it is read as another person (medium), never as a member.
6. The act's checked Q&A labels are tried before any of these rules.

Labelled Q&As

Labels that stage 2 missed are recovered first: initials it set aside as metadata, a label followed by a semicolon, initials run into the first word, a label written with another separator, and a guest's label run into the interviewer's line. Labels are then read in this order of precedence:

1. Q, Interviewer, Host, Reporter and similar, or a host named in a "NAME, HOST:" line: the interviewer, high.
2. A, Interviewee, Guest, Respondent and similar: the interviewee (below), at most medium.
3. Audience or caller: another person, high.
4. The act's checked labels. Those that name nobody on the line-up, such as an interviewer's initials, are another person (high).
5. A full name, alias or spelling variant: that person, high. The act's own name is read as its singer only in a record about the act.
6. A label of several words: a first name and surname, high; the first words of a member's name, such as "Slim Jim", under the rules above; otherwise another person, high.
7. A label of one word: initials, then a surname, then a first name, under the rules above; otherwise another person, medium.

The interviewee is taken from the title and an introduction of up to 2,500 characters before the interview. The first test that fits decides:

1. The person the title is about, named at its head or cued, as in "Interview with X", "X talks", "X on" or "Y interviews X": medium.

2. Where the title is about the act: a member the introduction presents as speaking (medium); otherwise a member who dominates the title and introduction, with three or more mentions there, twice the next member's and more than any non-member's (low); otherwise "band member, unspecified" (low).
3. Someone the introduction presents as speaking: medium.
4. The only person the title mentions, if a member, or a member with three or more mentions and 1.5 times the next member's: low.
5. Otherwise unknown. Authors, hosts and named journalists are never the interviewee.

The interviewer. A label that names nobody is the interviewer's when it is the author's name or initials (high), the outlet's name or initials (medium), or when at least half of its three or more lines end with a question mark (medium). An unlabelled question goes to the interviewer (high) when the next paragraph carries another speaker's label; otherwise it stays with the current speaker, so a question just before the interviewer's own label is the guest asking back. Where only the answers carry labels, a paragraph of the guest's answer that contains a question mark stays the guest's when more of the answer follows without a label, provided it is in the first person, is not put to "you" and does not name the guest.

Several interviewees. Where the title or introduction names two or more people and the answers carry no labels: "band member, unspecified" (medium) when all are nomination-era members, "other-era member, unspecified" (medium) when all are from other eras, ambiguous (low) when the eras are mixed, and unknown when any of them is not a member.

Answers printed without a label. Straight after the interviewer's question, request or short remark to "you" (up to 250 characters), a paragraph that reads as speech and does not name its speaker goes to the one person the other labels name, otherwise to the interviewee (at most medium); to "band member, unspecified" (low) if several members answer; otherwise it is unknown. This rule is off in radio, caption, lecture and one-sided transcripts.

Labels that are not speakers. A fixed list of 392 labels and a set of field names, such as photo credits and release details, are never speakers, except that initials on the list, such as PR, are a speaker's label when someone named in the record has them. Lines with field names are non-speech; page furniture is non-speech; a short line that looks like a heading is a heading. Text before the first label is narration (medium), the interviewer's if it is a question, and unknown if it is in the first person.

Caption transcripts. The host's and the guest's turns come from the likeliest alternation of the two over the whole programme. Where there is an audience, the guest gets only the answers to questions, at low confidence.

Unlabelled Q&As

1. **Questions.** A question is a paragraph of 8 to 600 characters that ends with a question mark (ignoring a closing bracket or quotation mark), does not open with a quotation mark, is not a short list item and is less than 70% capitals. A question to "you" that runs into a first-person answer is split at the answer, and prompts or statements followed by a first-person answer are the interviewer's too. The scraped text keeps no bold, so formatting plays no part.
2. **Answers.** An answer runs from its question to the next question, to page furniture, or to a narration paragraph: a quotation attributed in the third person, or 400 or more characters with no first person. Text outside the questions and answers goes to stage 4 as prose.
3. **Whose answer.** The interviewee's, or a member's whose label starts or interrupts the paragraph. On every route, an answer of 80 or more characters with no first person that names the interviewee becomes unknown, since it reads as narration about them.
4. **Confidence.** Questions are high; answers take the interviewee's confidence: medium, or low for the weaker tests.

First-person accounts

The first test that fits decides whose account it is:

1. **The member's own website,** where the site's name is the member's name, perhaps with "the" or "official" before it or "music", "official", "band" or "hq" after it: that member, medium.
2. **A narrator who names themselves** in the first 300 characters ("My name is", "I'm", "I am" or "This is" and a name) and is not a member: that person, high. A member naming themselves goes on to the next test.
3. **The interviewee,** where the title or introduction presents the piece as theirs, as in "Soundtrack of my life", "X on Y", a quotation followed by a name, or a playlist: medium. A name merely at the head of the title, as in an obituary or a fan's tribute, does not count.
4. **The act's own website:** the solo artist (medium), or "band member, unspecified" for a group (low).
5. **The author:** another person, medium.
6. Otherwise unknown.

Inside an account, a section heading that names a member starts that member's turn (medium); a heading that names anyone else starts another person's. Text before the first speaker heading is narration, and on pages whose title or address says interview or Q&A, questions to "you" are the interviewer's.

Speaker types from stage 3

Band member and "band member, unspecified" are band voice. The other types are other-era member, "other-era member, unspecified", interviewer, other person, ambiguous and unknown.

Stage 4: quotations in prose

Stage 4 finds the quotations in prose and decides which are speech. It reads the prose records and the prose introductions and tails of Q&A and first-person records; prose between two blocks of a Q&A is read as part of the interview instead (stage 5). A paragraph here is one line of the text.

The quotation convention

British publications usually put speech in single quotation marks and a quotation within a quotation in double ones; American publications do the reverse. An extractor that looks only for double marks misses most of the speech in a British-style record and picks up quotations within quotations instead. So the convention is set for each record by counting attributions beside each kind of mark: closing marks after a comma, full stop, question mark, exclamation mark, ellipsis or dash that are directly followed by an attribution, as in *'It's all happened so fast,' he admits*. Single marks inside a pair of double marks are ignored, since they are quotations within quotations. A record is British when single-mark attributions outnumber double-mark ones and there are at least two; otherwise it is American.

Quotation marks and apostrophes

With straight quotes, a single quotation mark is the same character as an apostrophe. A single mark between two letters or digits is an apostrophe, and elisions ('em, '80s), plural possessives (unless a pronoun or "says" or "said" follows) and the 'n' of rock 'n' roll are not quotation marks either. If a paragraph has an odd number of double marks, an inch mark after a one- or two-digit number is set aside.

Pairing

Only the marks of the record's convention are paired, in order, within a paragraph.

1. A quotation left open at the end of a paragraph carries on when the next paragraph opens with a quotation mark, as a long quotation over several paragraphs does.
2. It also carries on over up to 1,500 characters of paragraphs with no marks when the next mark it meets is a closing one, as on pages broken into short printed lines.

3. Otherwise the whole paragraph becomes an unresolved span. Its quotations are dropped and it belongs to no turn, so it reaches neither layer. If a quotation carried forward never closes, its first paragraph is unresolved and the rest is read again.
4. A quotation over 1,500 characters flags the record as probably missing a closing mark.

In all, 2,682 paragraphs are unresolved.

Speech, non-speech or undecided

Each quotation is read with up to 160 characters on each side, within its paragraph and not past the next or previous quotation. The first test that fits decides:

1. A singing verb after it: non-speech (lyrics).
2. An attribution straight after it, as in *"...," she says*: speech.
3. An attribution straight before it, with up to nine words after the verb, unless those words end in a title word: speech.
4. A sentence that opens with its speaker and a speech verb (or believes, thinks or feels) and ends in a colon just before the quotation, with at most 140 characters between the verb and the colon: speech.
5. An inverted attribution, as in *Says Mike,*: speech.
6. A singing verb before it: non-speech.
7. A participle, as in *, saying:*: speech.
8. A dash and a name after it, when the quotation is over 20 characters and not in title case: speech.
9. A title word beside it, such as called, dubbed, so-called or lyric: non-speech.
10. Lines run together with slashes: non-speech, since they are lyrics.
11. Nothing but spaces, line breaks or one punctuation mark since the previous speech quotation or its attribution: speech, since it continues.
12. Title case, at most ten words and no sentence break: non-speech.
13. A word in the first or second person, at least four words and a word that reads as a verb: speech.
14. Otherwise undecided.

One test overrides this order: an attributed quotation of at most eight title-case words, with no first or second person, whose clause says wrote, recorded, released or similar, is non-speech even when an attribution has made it speech, since it is a title. A "NAME:" at the start of the paragraph, within 80 characters of the quotation, labels it, and a labelled quotation that would be undecided becomes speech when the label names a member of either era or the band. An undecided quotation stays inside the narration turn, which is

flagged, and never counts as band voice. There is no separate test for scare quotes, so a word in quotation marks to show doubt stays undecided unless a title word introduces it. Length is never used on its own: short speech, such as *"No," he says*, is real.

Stage 5: who speaks a quotation

Each speech quotation in prose goes down a cascade of tests, and the first that fits names the speaker and sets the confidence. The Q&A and first-person blocks never enter it, since stage 3 has named their speakers. The speaker is read from the attribution, cut at a sentence break; in a clause such as *, saying:*, it is the last person in the sentence who is followed by a verb, otherwise the last he, she or they.

Two terms recur. **The title or introduction** means the title and the first 1,500 characters of the text. A member is **mentioned** when named in full, or by a first name of four or more letters or a surname of three or more that is not an ordinary word; a name that several members share counts for the one most recently named in full.

The tests, in the order the code applies them, with the rule each is recorded under in the turns table:

1. **A pronoun** (rule 5). He, she or "the singer" goes to the current speaker (below); they or we only when the current speaker is the band or "band member, unspecified". "I" in the narration is the journalist (medium).
2. **A relation and one name,** as in "his wife Jane" (rule 2): another person, low.
3. **A member's name** (rule 1). A member's full name or alias, one member's exact full form, or a first name and surname that fit only one member: that member, high.
4. **The act's own name.** For a solo act, the artist (rule 1, high); for Sade, Sade Adu (rule 1, medium), since the name is both the band's and the singer's; for a group, the band as a group (rule 4, medium).
5. **Another name of two or more words** (rule 2): another person, high when the first word is a listed first name, otherwise low.
6. **One word that only one member has,** and that no non-member in the record shares (rule 1): that member, high, when the record mentions the member; otherwise ambiguous, low. A name that is also an ordinary word, such as Day, is not read as a name at all.
7. **One word that several people share** (rule 3). First, the person the title or introduction is about (medium); otherwise the candidate mentioned last before the quotation (medium); otherwise "band member, unspecified" (medium) when every candidate is a nomination-era member; otherwise ambiguous (low). "Cherry says" in a piece titled "Neneh Cherry resets her career" goes to Neneh rather than to Don Cherry, named a paragraph earlier.

8. **One word that one non-member in the record has in full** (rule 2): that person, high. Any other capitalised word: another person, low.
9. **"The band" or "the duo"** (rule 4): the band as a group (medium), only when the act is not a solo act, the phrase starts with "the", any word for the group's size matches the nomination-era line-up, and the act is in focus. That means the act or a member is named in this paragraph or the two non-empty paragraphs before it (or in the title), and the last subject of a verb outside quotations is the act or a member, not another group. Otherwise, and for descriptions such as "a spokesperson", another person, low (rule 2). A member named beside the band, as in "the duo's Dave Stewart", is the speaker (high). A quotation given to the band becomes non-speech when it is two words or fewer, in title case, mostly capitals or run together with slashes, unless the attribution splits it.
10. **Rules 5b and 5c** (below), for a he or she that the current speaker cannot take, and for a description when there is no current speaker.
11. **Nothing fits** (rule 6): unknown.

A quotation with no attribution at all continues the current speaker (rule 5). With no current speaker it goes to rule 5c; failing that, it is unknown.

Rule 5, the current speaker. A pronoun goes to the current speaker when it does not contradict that speaker's gender: medium when the gender was checked, low when it could not be. Neither a pronoun nor a continued quotation is ever surer than the speaker it continues.

Rule 5b, the person the narration is about. A he or she that the current speaker cannot take, because there is none, the gender does not fit or the current speaker is the band, goes to someone named outside quotations since the last speech quotation, within this paragraph and the two before it, whom the pronoun fits: the latest followed by a verb, otherwise the latest who opens a sentence, otherwise the latest. A member gets medium, anyone else low. Without this rule, the quotation in *Knopfler wrote the song in 1978. "It came out of nowhere," he says.* would be unknown.

Rule 5c, the member the piece is about. Failing that, a pronoun that fits, or an unattributed quotation, goes to the member the title or introduction is about, at low confidence.

Q&A passages

Prose that lies between two blocks of a Q&A is read as part of the interview rather than by the cascade: 582 passages in all. Question lines are the interviewer's (medium), and the rest goes to whoever answered before, at most medium, and low for a member's answer that goes on without the first person. These rules are recorded under names that begin "Q&A passage", and they give 307 band-voice turns.

Rules added during development

Rules 5b and 5c and the first step of rule 3 were added after the development checks showed missed speech, and each was measured: in the development samples, rule 5b was right for 19 of 21 members and rule 5c for 13 of 19 cases. Each has a rule name of its own, so its turns can be set aside in a robustness check, with two cautions: a quotation that continues a 5b or 5c speaker carries a rule 5 name, and a joined turn shows only its first quotation's rule. Rule 5c's turns, and those that continue them, are low, so they are already outside the main set.

Two further changes adjust existing rules and have no rule names of their own: the gender check for people outside the act and the conditions for clearing the current speaker (stage 6). A third gives low confidence to another person named by one word or by a first name not on the list, or only described, and shows in the rule's wording: one-word names and unlisted first names had been right 64% of the time, unnamed speakers 0%.

Stage 6: the current speaker

The script keeps a current speaker as it reads each article, which is what lets rule 5 give a quotation with no name beside it to the right person. The risk is carrying a speaker too far, so the current speaker is cleared on clear signals.

1. **Set.** Any rule that names someone sets the current speaker, and so does a second-hand sentence (stage 7). An unknown or ambiguous result clears it.
2. **Distance.** It is cleared before a quotation that comes more than two paragraphs after the last speech quotation or second-hand sentence, counting lines of the text, blank lines included.
3. **A change of subject.** The script looks at the people named outside quotations since the last quotation. It leaves out the current speaker, names in this quotation's own attribution, the rest of the previous attribution's sentence, as in *"…," Stewart tells Billboard, referring to his and Annie Lennox's first appearance*, and names in passing, such as "the late John Lennon" or "John Lennon's son". If anyone remains:
   a. after "he" or "she", the current speaker is cleared only if one of them fits the pronoun;
   b. after any other attribution, it is cleared;
   c. before a quotation with no attribution, it is cleared only if one of them opens a sentence or is followed by a verb.
4. **Gender.** Each member's pronoun comes from the name lists. For anyone else it comes from a title (Mr, Ms, Mrs, Miss, Sir) or a first name that is clearly a man's or a woman's; names used for both are left out.

Stage 7: building the turns

Spans become turns in reading order, and each turn gets its position in the record, so the sequence of an article, and later of an oral history, can be rebuilt.

1. **Speech turns.** A speech span joins the turn before it when that turn is prose speech by the same speaker and only attributions or gaps without words lie between, as in *"We were exhausted," she says. "We had been touring."* Otherwise it starts a new turn. A joined turn keeps its first quotation's rule and confidence. The attributions are stored as its evidence and become a single space in its text.
2. **Narration turns.** The journalist's narration, with any non-speech and undecided quotations and the headings inside Q&A blocks, forms the journalist's turns, at high confidence. The journalist is the article's author when that reads as a person's name, otherwise the outlet. A narration turn that holds an undecided quotation is flagged.
3. **Embedded quotes.** Inner marks of the other kind around three or more words, as in *"He told me, 'You'll never make it,'" she recalls*, are someone else's words reported by the speaker. They stay inside the turn, which is flagged.
4. **No turn.** Unresolved spans, sung lines and stage directions belong to no turn.

Second-hand turns

A narration sentence that reports a member's words without quotation marks becomes a turn of its own. In the invented sentence "Adam says they have been practising for 10 hours every day", Adam's words reach the reader second-hand, whereas *"We have been practising for 10 hours every day," said Adam* is band voice. The search runs only in narration, and three patterns count:

1. A sentence whose subject, within its first 80 characters, is followed (allowing an aside and up to three adverbs) by a verb of reporting such as says, told, recalls, admits, claims, reveals, mentions, announces, denies or praises, with no double quotation mark in the rest of the sentence. "Maintains" counts only before "that", and "writes" only before "that", "on" or "in a post, statement or book"; "says nothing" does not count.
2. "According to" followed by a name, "the band" or "the group".
3. A tag such as *, he says,* in the middle of a sentence, at its end or after a dash.

The subject must resolve to a member of either era or to the band: a name through rules 1 to 4, "the band" through rule 4's tests, and "he" or "she" to the current speaker only if that speaker is a member of that gender (at most medium). If the current speaker does not fit, no turn is made; if there is none, rules 5b and 5c apply. The turn is the whole sentence and takes the confidence with which the member was identified. Its speaker is the journalist, the member goes in the reported member field, and it is never band voice. The member then becomes the current speaker.

Stage 8: interview dates

The oral history should be ordered by when each thing was said, which can differ from when the page was published. A 2012 retrospective may carry a singer's words from last month and also from a 1983 interview, so the interview date belongs to the turn, not the article. The rule is deliberately narrow, because a looser one had already failed in the date cleaning of Step 1, where phrases such as "first appeared in 1989", about a song, were taken for the date of the article.

1. **Prose.** A speech or second-hand turn takes the first year (1950 to 2029, and not after the publication year) in its attribution's sentence that fits one of three patterns: a speaking verb, up to five words, then "in" and the year, as in "told NME in 1983"; "in a 1985 interview"; or "interviewed" followed within 40 characters by "in" and the year, as in "Interviewed by Classic Rock in 2019". A quotation with no attribution takes no stated year unless it is joined to one that has it.
2. **Q&As and first-person accounts.** The title, or the 1,500 characters before the first block, can date the whole interview, as in "The 1982 Stray Cats Interview" or "This interview was conducted in 1992". That year then applies to every turn of the blocks, questions included.
3. **Otherwise,** and for all narration, the publication date is used. It is also used for anything looser, such as a death year, a recording year or "our 2014 interview" in a 2022 piece.

Each turn stores its interview date, the source (stated in the text, or the publication date as a proxy), the precision and the words the date came from, in columns of their own, so the record's date is never overwritten. 1,789 turns have a stated year. The turns table writes a stated year as the year alone; the oral histories date it to 1 January of that year, with precision "year", so that it sorts.

Stage 9: confidence and the main set

Each turn's confidence is a word, not a number, since a figure such as 0.99 would imply a measurement that has not been made. It starts at the level of the rule that named the speaker:

How the speaker was named	Confidence
Q&A label: a full name or alias, an interviewer's label such as Q or Host, or the author's name	High

How the speaker was named	Confidence
Q&A label for the interviewer taken from the outlet's name or from its questions	Medium
Q&A answer or first-person account, the speaker taken from the title, introduction or a heading	Medium; low for the weaker tests
Q&A passage: a question line	Medium
Q&A passage: an answer	At most medium; low for a member's answer without the first person
Rule 1: a member's name beside a speech verb	High (Sade Adu, from the act's name: medium)
Rule 2: another named person	High; low for one word that nobody in the record has in full, a first name not on the list, or a description
Rule 3: a shared name settled by the title or introduction, or by the last mention	Medium
Rule 3: every candidate a nomination-era member ("band member, unspecified")	Medium
An ambiguous name (rules 1 and 3)	Low
Rule 4: the band named as the speaker	Medium
Rule 5: a pronoun, or continuing the current speaker	Medium; low without a gender check
Rule 5b: the person the narration is about	Medium for a member, low for anyone else
Rule 5c: the member the piece is about	Low
Rule 6: nothing fits	Unknown
A second-hand turn	That of the member's identification
The journalist's narration	High

The level then drops for known risks:

1. One step, never below low, when the member was named by a single word and a non-member named anywhere in the record shares the member's surname.
2. A turn is never surer than the speaker it continues.

The band as a group stays at medium even when the attribution is explicit, for three reasons. A group statement is often a press release written on the band's behalf, so it is the band's position but not any member's own words. "The band" can mean another act in the same article. And several act names are also ordinary words or places, such as Asia and Big Country, or name a member as well as the group, as Sade does.

The main set is the band-voice turns at high or medium confidence: 16,870 of the 19,713. No turn is dropped for its confidence. Low turns stay in the data with their level, outside the main set, and a set of high turns alone serves as a robustness check.

Output

The last script writes three linked tables, joined by Record ID: as JSON with every text whole, and as Excel for reading, with a dashboard.

The turns table has one row per turn, 132,835 in all, in 37 columns:

1. Record ID, Turn ID, Turn order and Turn type (speech, narration or second-hand).
2. Speaker type, Speaker name, Speaker candidates (where a name was ambiguous) and Reported member (second-hand turns only).
3. Band voice, Main set (band-voice turns only), Confidence, Attribution rule and Attribution evidence.
4. Embedded quote, Question (question lines in Q&A blocks, though not every interviewer line of a labelled Q&A is marked) and Holds an undecided quote.
5. Turn text (the words verbatim with their quotation marks, leaving out the attributions between the parts and the Q&A labels), Words and Repeat of.
6. Span start and Span end, the character positions in the stage 1 text.
7. Interview date, with its source, precision and evidence.
8. Route, Quotation convention, eight fields carried over from the record (such as the act, the outlet and the publication date), Rules version and Human verified.

Speaker types are band member; band member, unspecified; band collective; other-era member; other-era member, unspecified; interviewer; journalist; other person; ambiguous; and unknown. Band voice is a speech turn by one of the first three.

The spans table, in JSON only, holds every span: Record ID, Span order, Span type (speech, narration, second-hand, attribution, non-speech quote, undecided quote, unresolved, label, sung, stage direction or page furniture), Span start, Span end, the Turn ID it belongs to (blank for a span in no turn) and the evidence for its type.

The records table, also a sheet of the dashboard, keeps ten columns of the cleaned corpus and adds the route and its evidence, whether the record was processed, the quotation convention, counts of attributions by kind of mark, counts of spans, turns,

speech, band voice, main set, second-hand, unknown and unresolved, the round trip and any flags.

Repeats. A turn of eight words or more whose text matches an earlier turn of the same type gets that turn's ID in Repeat of. The comparison ignores case, spacing, the style of quotation marks and punctuation at either end, and "earlier" means by interview date, then publication date, then record and turn order. 9,662 turns repeat an earlier one, mostly in republished articles. They are marked, not removed.

What the tables hold

1. **Turns:** 132,835: 86,870 of speech, 44,065 of narration and 1,900 second-hand.
2. **Band voice:** 19,713 turns and 1,451,057 words in 2,447 records. By speaker: a named member 19,531, "band member, unspecified" 157 and the band as a group 25. By confidence: high 8,060, medium 8,810 and low 2,843.
3. **The main set:** 16,870 turns and 1,290,055 words.
4. **Band-voice turns by route:** prose 8,774, labelled Q&A 6,229, unlabelled Q&A 4,534 and first-person 176.
5. **Outside band voice:** 687 speech turns by other-era members and 8,716 speech turns whose speaker is unknown or ambiguous.
6. **Dates and gaps:** 1,789 turns carry a stated interview year, and 2,682 paragraphs are unresolved.

Step 2 then builds one oral history per act from these tables: every turn of every article, oldest first, each with its voice (band, public narrative, second-hand, other-era member or unknown), with later copies counted rather than kept. The technical appendix describes them.

Validation

The rules were developed on some records and measured on others. Blind checks by AI agents guided development; a person's coding of held-out records measured the result once, and its misses were not fixed, so those records were never used for tuning.

Checks built into the scripts

1. **The round trip.** Every record's spans must rebuild its normalised text exactly, with no gaps and no overlaps. The records table reports the result for each record, and no record fails.
2. **Expected inputs.** Each script checks that the files it reads are the expected versions, by their record or route counts and, for the cleaned corpus, a fingerprint of its record IDs and addresses, and stops if they are not.

3. **The same result.** Stages 4 to 9 compare their counts and a fingerprint of the turns with the checked run, and the dashboard's Reproducibility line reports whether they match; stages 1 to 3 compare their counts. During development, two runs with different Python hash seeds gave identical files.
4. **Named records.** After a run, two test scripts check the records named in the rules: `tests/s3_tests.py` (68 checks) and `tests/s49_tests.py` (45 checks, four of them on the whole table).

Checks during development

The routing check was made by hand; the others were made blind by AI agents (Claude), which never saw the script's answers.

Check	Sample	Result
Routing (stage 2), by hand	54 records	The first rules agreed on 41 of 54. The second version, revised on these same records, agrees on 45 and fixes all 9 misplaced block starts
Routing, by AI agents	120 and 60 records	Routes agreed on 109 of 120 and 57 of 60
Q&A speakers (stage 3)	80 fresh turns	39 of 40 band-voice turns confirmed; none of 40 other turns was a member's
Q&A answers read as questions	1,268 Q&A turns: 1,044 interviewer turns and 224 band-voice turns	106 interviewer turns held a member's words; the rules were then changed to fix those that the labels make certain
Prose quotations (stages 4 to 9)	144 records, 1,958 quotations; 32 records held out	Held out: precision 100% (64 of 64), recall 94% of marked member quotations (about 80% counting unpaired marks)
Confidence levels	The same 144 records, 112 of them used in development	Speaker right: high 96%, medium 96%, low 73% (targets 95% and 80%)

Validation against a person's coding

1. **Held out.** Every record used in development, 1,314 in all, was set aside: records coded earlier, every record in the agents' checks and the routing review, records named in the rules and tests, records in every review sheet, and 99 more that share 10% or more of their words with them. The draw came from a pool of 7,856 records.
2. **The draw.** 40 records, stratified by route and by whether the script found band voice (7 with it and 3 without, per route). Two prose slots were kept for British-style records and one for Julian Lennon. The draw (seed 20261025) also had to include collision acts, acts with relatives on the lists, and thin and rich acts, with no act more than three times and no outlet more than five.
3. **Coding.** Units were sentences, also cut at quotation marks, worked out from the text alone. Long records (over 2,500 words) were coded in an opening of about 300 words and a window of about 2,000. Nothing from the script's output was in the coding workbook, and the records were in random order. The codes were each record's members and other-era members, a member not listed, the interviewer, the writer's narration, the writer reporting a member's words, another person, can't tell who, and not text.
4. **Scoring.** Character by character against the spans table, leaving out attribution tags, labels, whitespace and text coded as not text or as a member not listed. 21 of the 40 records were coded and scored.

Results.

1. **Band voice:** precision 96.4% and recall 82.9%, and the member was right in every case. In the main set: precision 96.8% and recall 79.9%. Four records were coded after feedback on the coding rules; without them, precision is 95.9% and recall 83.4%.
2. **By route,** precision and recall: prose 99.7% and 70.0%; labelled Q&A 96.4% and 93.5%; unlabelled Q&A 99.3% and 67.7%; first-person 92.9% and 100%.
3. **Confidence of band-voice turns:** high 42 of 42 right, medium 55 of 56 (targets 95% and 80%).
4. **Turn boundaries:** 95.0% of the script's speaker changes match the coder's, and the script finds 78.7% of the coder's.
5. **Second-hand turns** agree poorly: 190 characters in common, of 912 that the script marked and 669 that the coder marked (precision 20.8%, recall 28.4%).
6. **Formats:** the coder's format matches the route in 13 of 21 records. Stated interview years could not be tested.
7. **Causes of the missed characters:** a Q&A with a non-member guest, where the member's labelled answers were left unknown (62%); the surname Day (Valerie Day of Nu Shooz), read as an ordinary word (25%); an article read as a labelled Q&A (8%); and unattributed or partial quotations.

8. **Causes of the false alarms:** an interview read as a first-person account (61%); a producer's unattributed quotation given to the member by rule 5c (16%); interviewer prompts without a question mark (13%); and another writer's section in a first-person page (9%).

Limits of the validation.

1. The held-out records were coded by one person; agreement with a second coder has not been measured.
2. There is no corpus-level figure. Prose is 92% of the pool but has five scored records, and one of them would carry most of the weight.
3. The checks by AI agents do not replace the human validation; they show where the rules were tested before it.

Known limits

These limits are known and were left in place. Fixing those that the validation found would have tuned the rules on held-out records.

1. **Unattributed fragments,** such as *she called it "a mess"* with no speech verb, stay undecided and never become band voice. In the development checks they were the largest source of missed member speech in prose.
2. **One wrong guess carries on.** An unattributed quotation continues whoever spoke last, so a wrong guess by rule 5b or 5c carries through the quotations after it, at the same or lower confidence.
3. **Unpaired quotation marks.** 2,682 paragraphs whose marks do not pair are in neither layer. musicmusingsandsuch.com, for one, quotes long excerpts from other reviews over several paragraphs without reopening the marks.
4. **Inch signs at the end of a quotation.** Stage 1 turns a double quote straight after 7, 10 or 12 into an inch sign, so a quotation that ends with one of these numbers, as in *"...aged 12"*, loses its closing mark.
5. **Q&A answers read as questions.** Where a Q&A's questions carry no label, a guest's paragraph that ends with a question mark is still read as the interviewer's next question: about 210 turns, or 0.8% of the band-voice words. Rules tested on 1,268 blind-coded turns could not find these answers without also moving real questions, so only the cases that a page's labels make certain were fixed.
6. **Questions inside answers.** When an answer and the next question share a paragraph, the whole paragraph goes to one speaker.
7. **Q&As with a non-member guest** can leave the member's labelled answers unknown (5 records and 46 turns in the corpus). This was the largest cause of missed characters in the validation.

8. **Names that are also ordinary words,** such as Day for Valerie Day of Nu Shooz, are not read as names when they stand alone.
9. **Freddie Waite.** In Musical Youth's records the name usually means the father of the member Frederick Waite, who formed the band, but the son's alias makes the scripts read it as the member, at high confidence.
10. **Two aliases written with slashes,** "Susanne / Susan / Ann Sulley" and "Paul / Pål Waaktaar", never match, although the members' other forms still do.
11. **Headline pull-quotes,** where the title repeats a quotation from the body, are given to the member by rule 5c or left unknown.
12. **Records not in English.** Stage 2's tests for pronouns cover French, Spanish, German, Italian and Danish, but from stage 3 on the 28 records in other languages are read with English rules, so their quotations are mostly undecided.
13. **Pages with only single marks** and no "says" clauses are read with double marks, so their quotations stay in the narration. Reading them in British style would have left 705 paragraphs unpaired.
14. **A caption after an answer.** In a Q&A, a photo caption of seven or more words straight after a member's answer becomes part of that answer.
15. **Second-hand turns are found narrowly.** Against a person's coding, their precision was 20.8% and their recall 28.4%.
16. **Settings not tested.** The two-paragraph limit for the current speaker is the starting value the original plan set, and stated interview years could not be tested in the validation.